INTELIGÊNCIA ARTIFICIAL

Aprendizado Descritivo
Clustering

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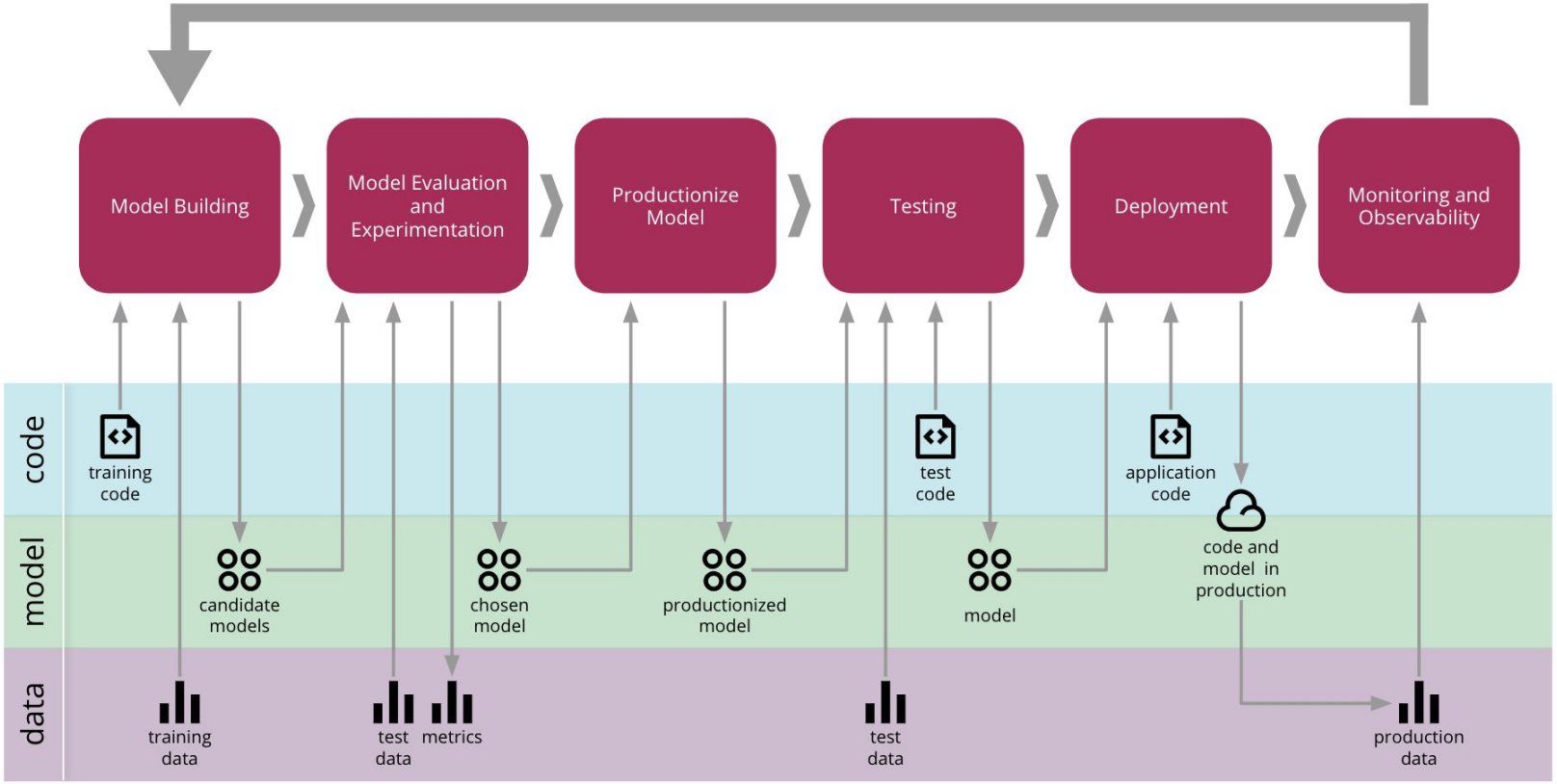
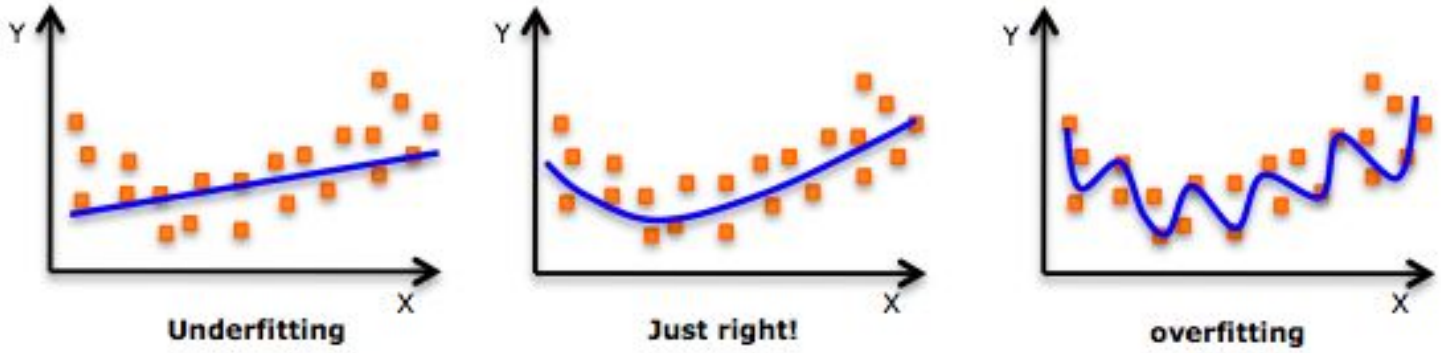
C.E.S.A.R school



Continuous Delivery for Machine Learning

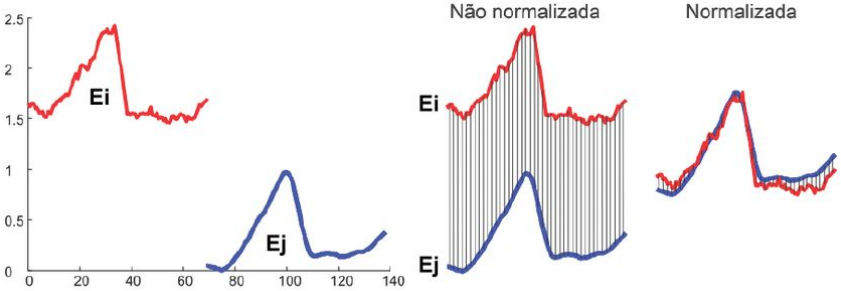
Processo de Validação

Como garantir que o modelo obteve um bom resultado?



An example of overfitting, underfitting and a model that's "just right!"

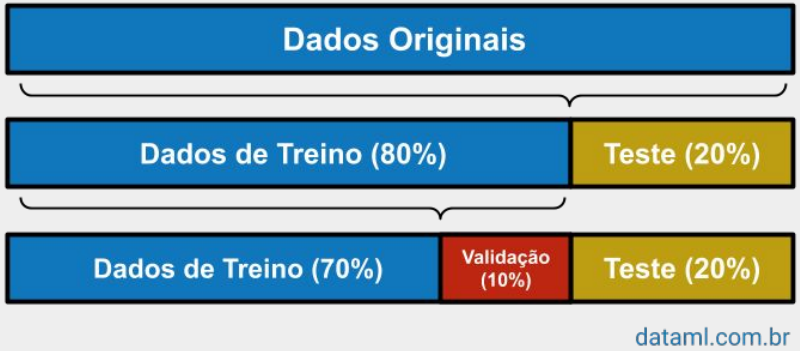
Preparação de Dados



Limpeza, normalização e filtros a partir do que faz sentido no problema estudado

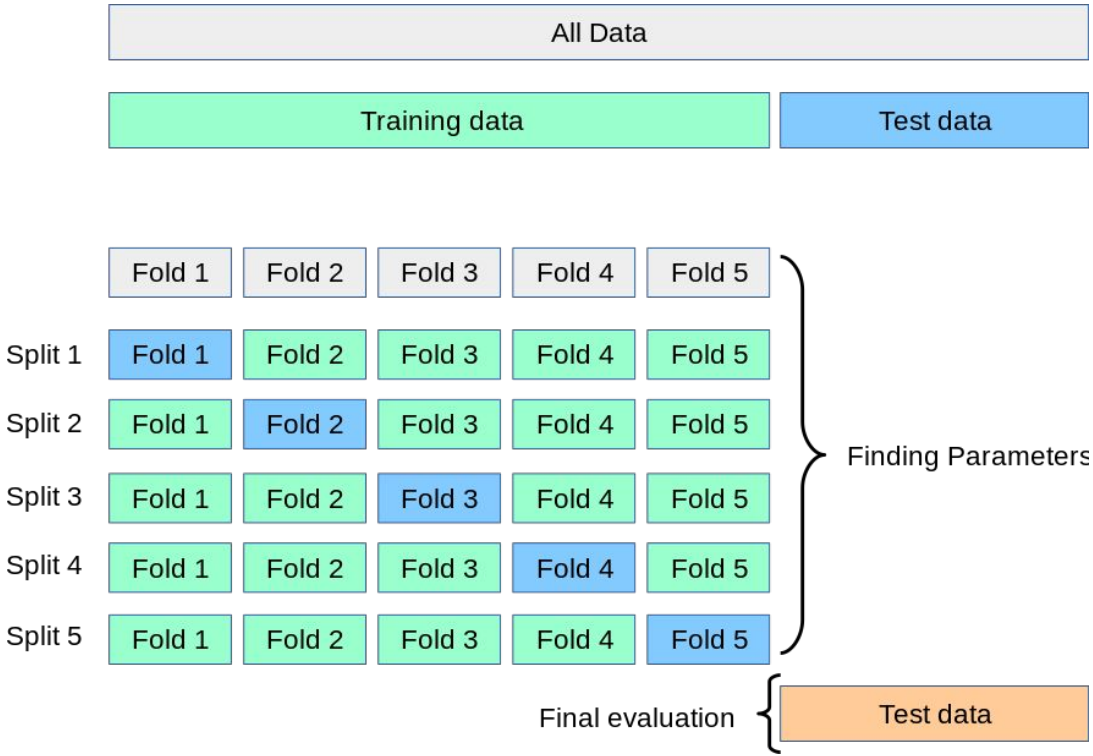
Etapa de Treinamento

Separação de dados



Processo de Validação

Uma forma de diminuir o desvio do erro é através do cross-validation



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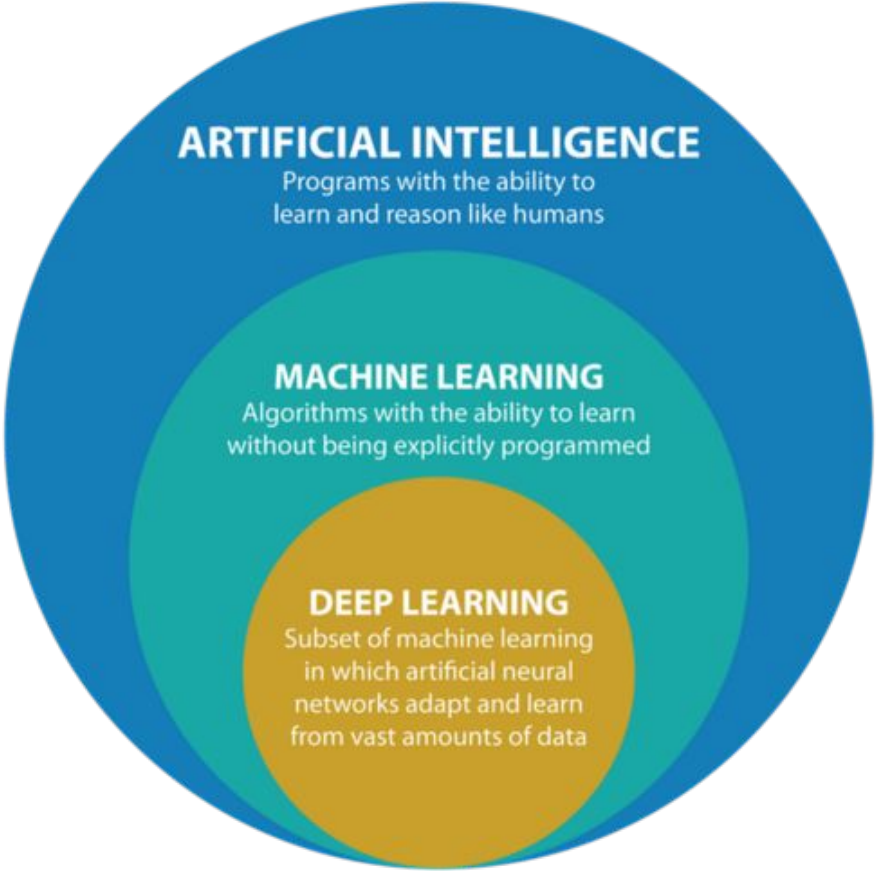
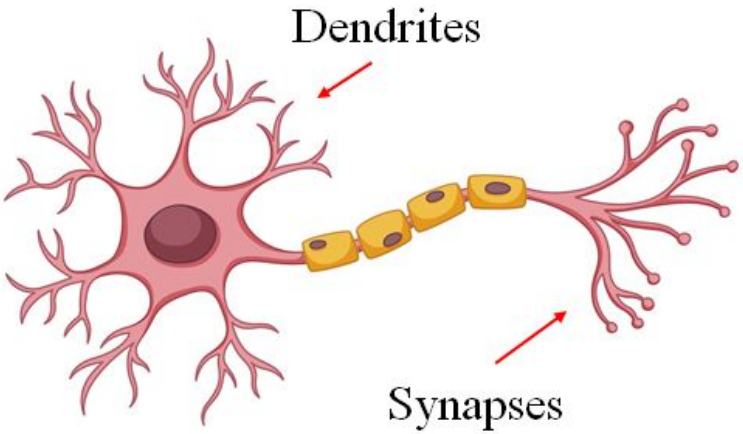
Redes Neurais Artificiais

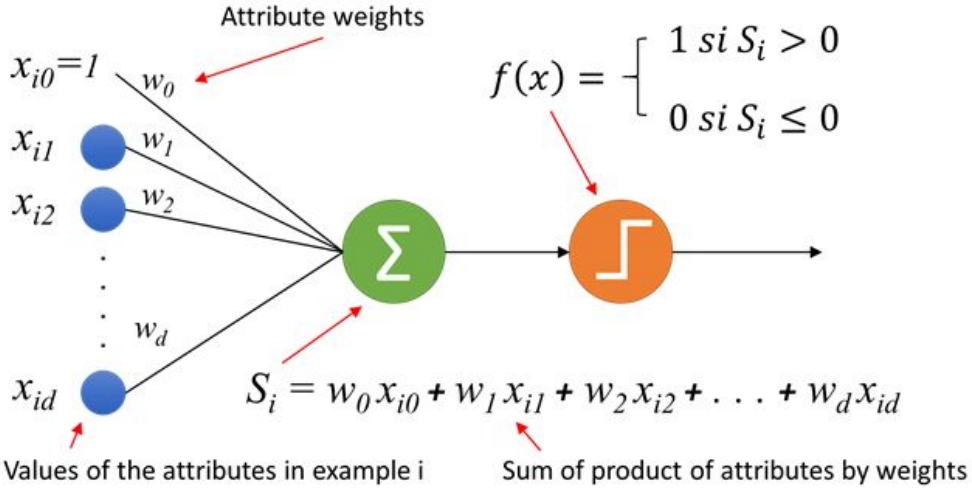
Machine Learning vs IA



Redes Neurais Artificiais - Perceptron

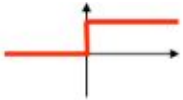
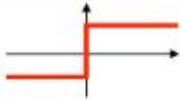
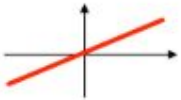
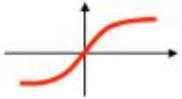


NEURON

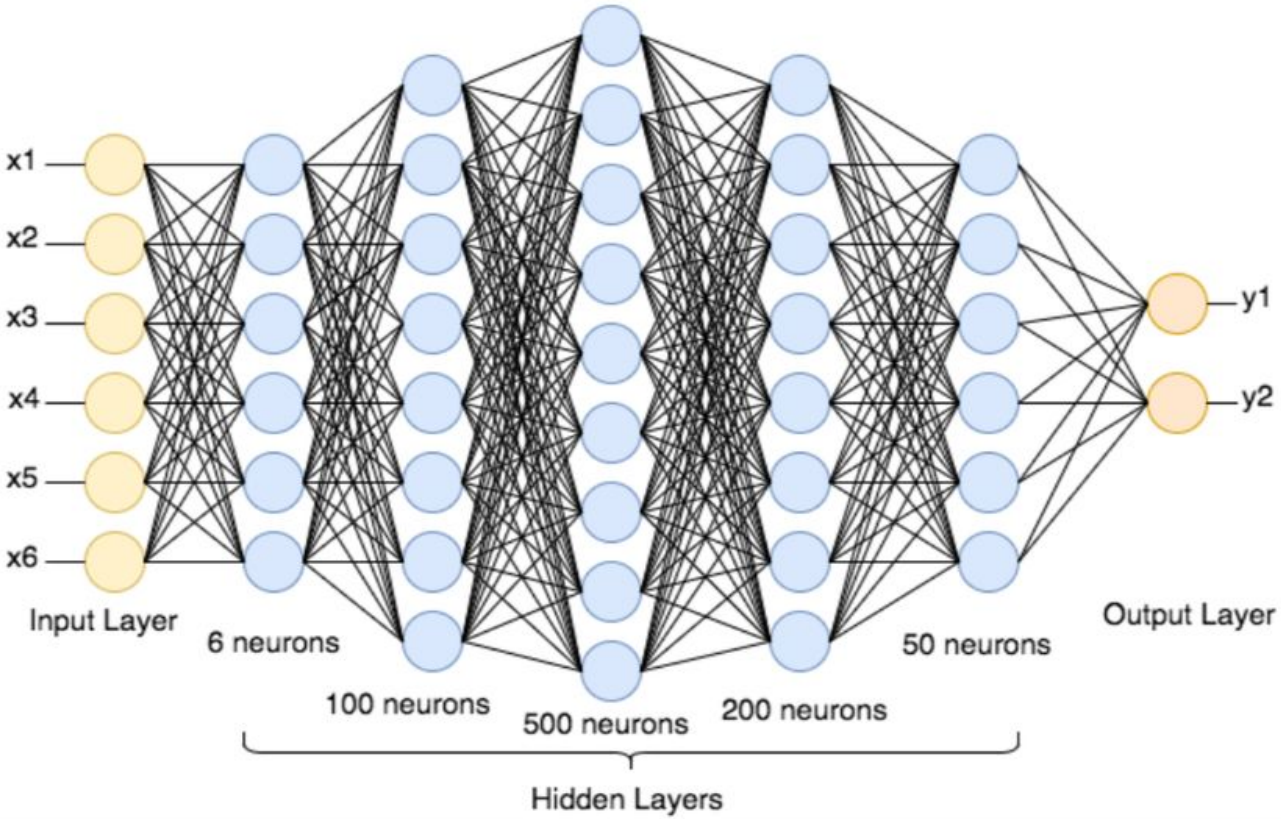


PERCEPTRON

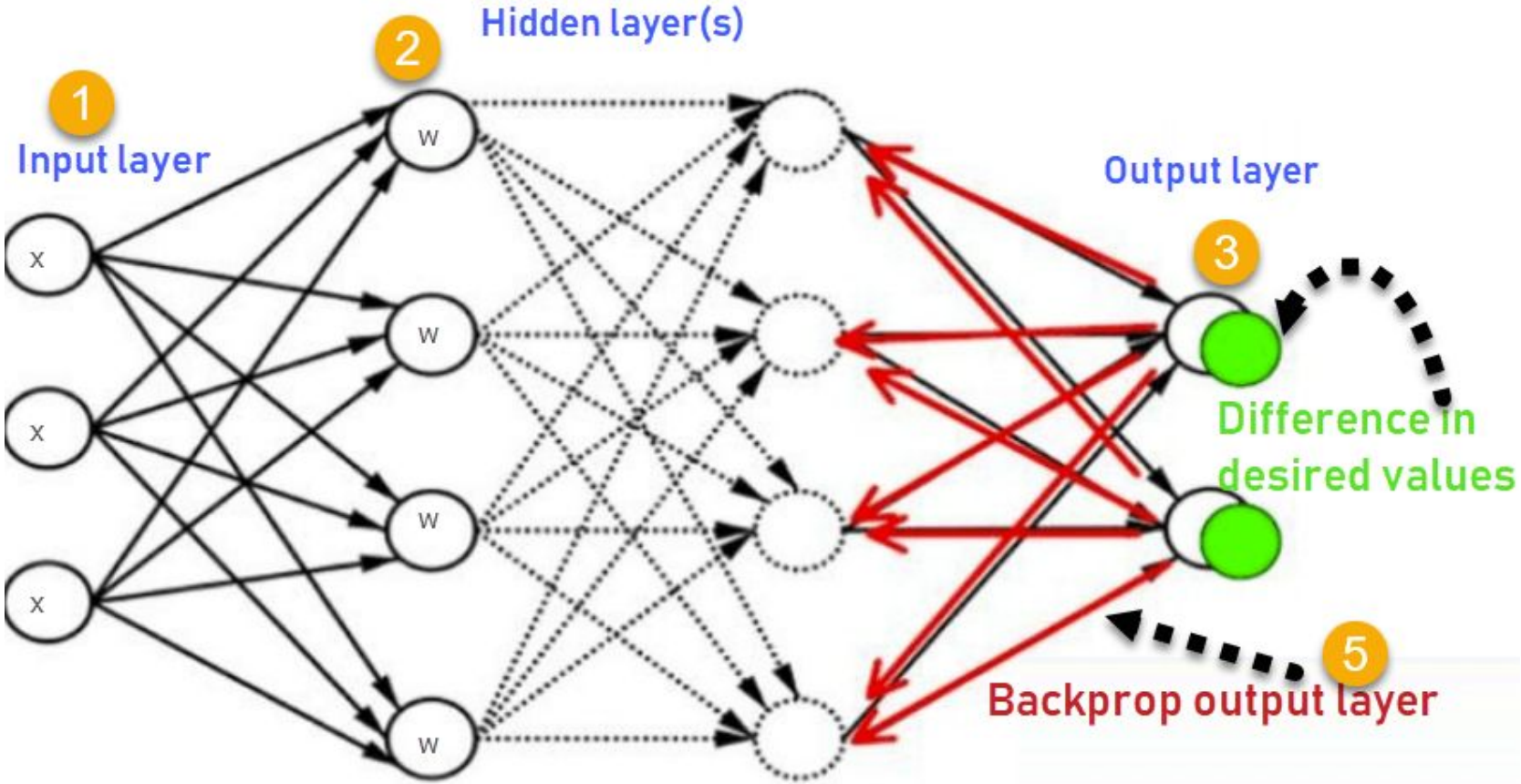
Redes Neurais Artificiais - Funções de Ativação

Unit step (Heaviside)	$\phi(z) = \begin{cases} 0, & z < 0, \\ 0.5, & z = 0, \\ 1, & z > 0, \end{cases}$	Perceptron variant	
Sign (Signum)	$\phi(z) = \begin{cases} -1, & z < 0, \\ 0, & z = 0, \\ 1, & z > 0, \end{cases}$	Perceptron variant	
Linear	$\phi(z) = z$	Adaline, linear regression	
Piece-wise linear	$\phi(z) = \begin{cases} 1, & z \geq \frac{1}{2}, \\ z + \frac{1}{2}, & -\frac{1}{2} < z < \frac{1}{2}, \\ 0, & z \leq -\frac{1}{2}, \end{cases}$	Support vector machine	
Logistic (sigmoid)	$\phi(z) = \frac{1}{1 + e^{-z}}$	Logistic regression, Multi-layer NN	
Hyperbolic tangent	$\phi(z) = \frac{e^{z} - e^{-z}}{e^{z} + e^{-z}}$	Multi-layer Neural Networks	
Rectifier, ReLU (Rectified Linear Unit)	$\phi(z) = max(0, z)$	Multi-layer Neural Networks	

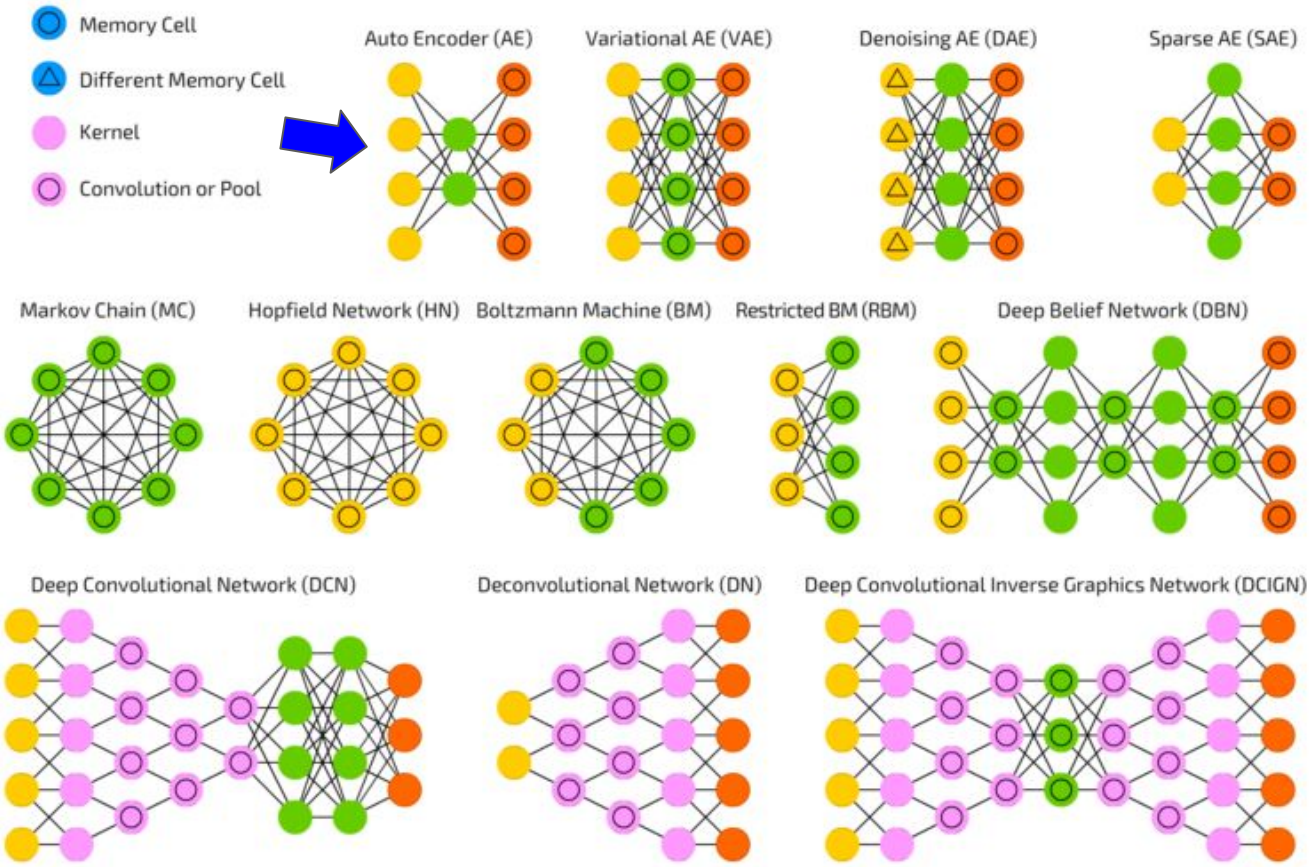
Redes Neurais Artificiais - Multi-Layer



Redes Neurais Artificiais - Backpropagation



Redes Neurais Artificiais - Arquiteturas



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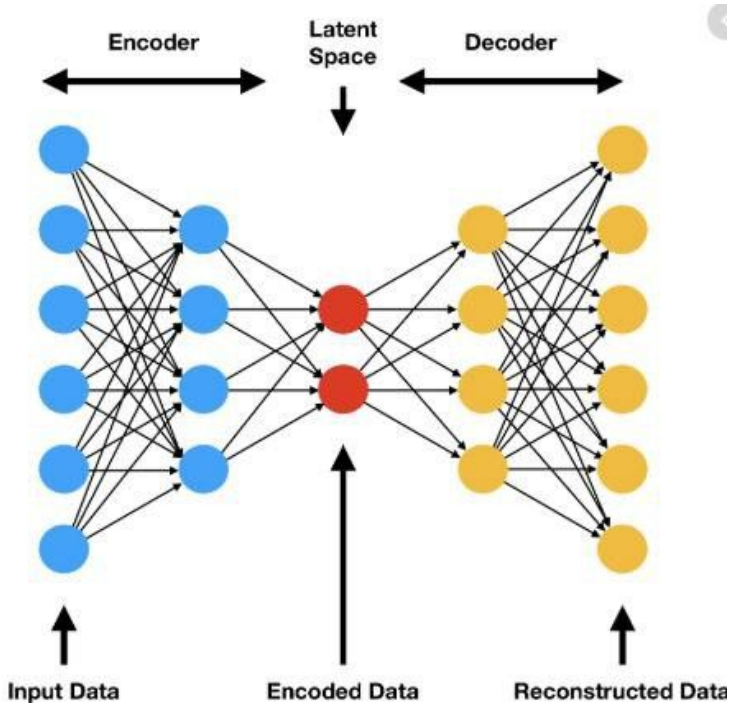
AutoEncoders

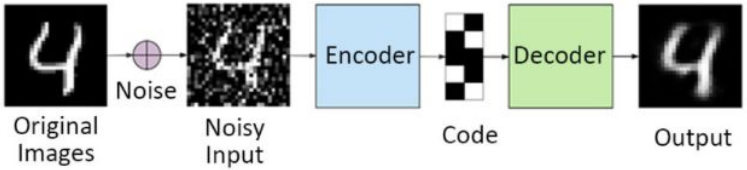
Partes do AutoEncoder

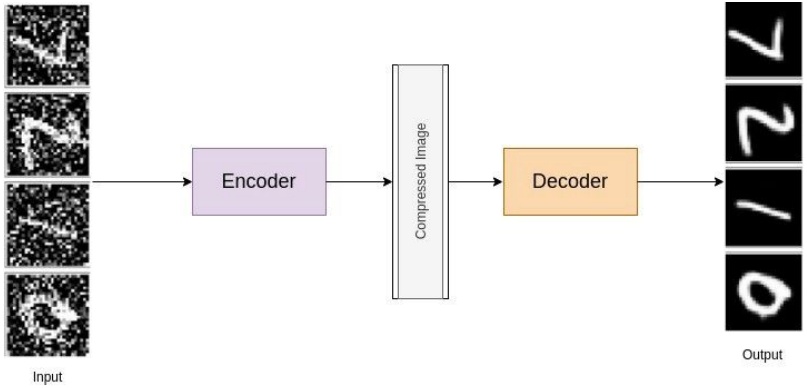
- O autoencoder é composto de 3 partes:
  - input: o array de características
  - espaço latente: representação dos dados em baixa dimensionalidade
  - output: reconstrução dos dados originais
- O autoencoder é realizado em duas etapas:
  - encoder: descoberta do espaço latente
  - decoder: reconstrução dos dados



AutoEncoder utilidades

- o autoencoder é utilizado para reconstrução de dados com ruído (eliminado ao longo das camadas);
- e também para aumento do conjunto de exemplos





Deep Learning - Libraries

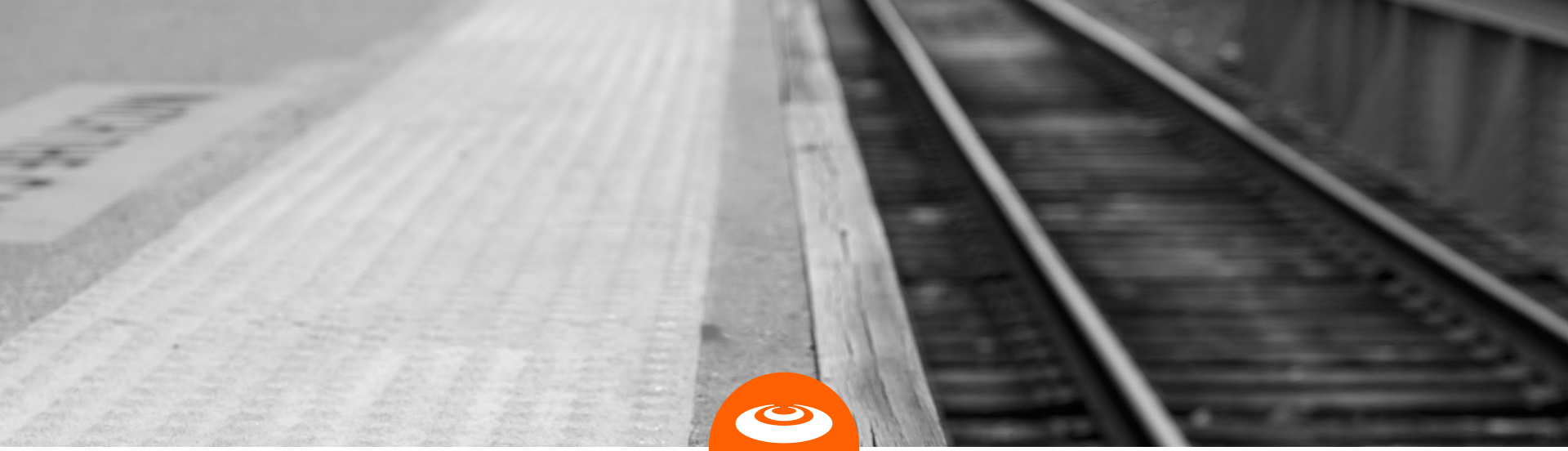
TensorFlow

Keras

PyTorch

scikit learn

python



Hands On!

C . e . S . A . R

Pessoas impulsionando inovação.

Inovação impulsionando negócios.

NOSSO CONTATO

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